Chapter-12

06.10.2026

Measure of Dispersion

- Data কতটা ছড়িয়ে আছে তার হিসাব।

Ex:- • 48, 49, 50, 51, 52, mean $= \frac{48+49+50+51+52}{5}$

+2

$= \frac{250}{5} = 50$

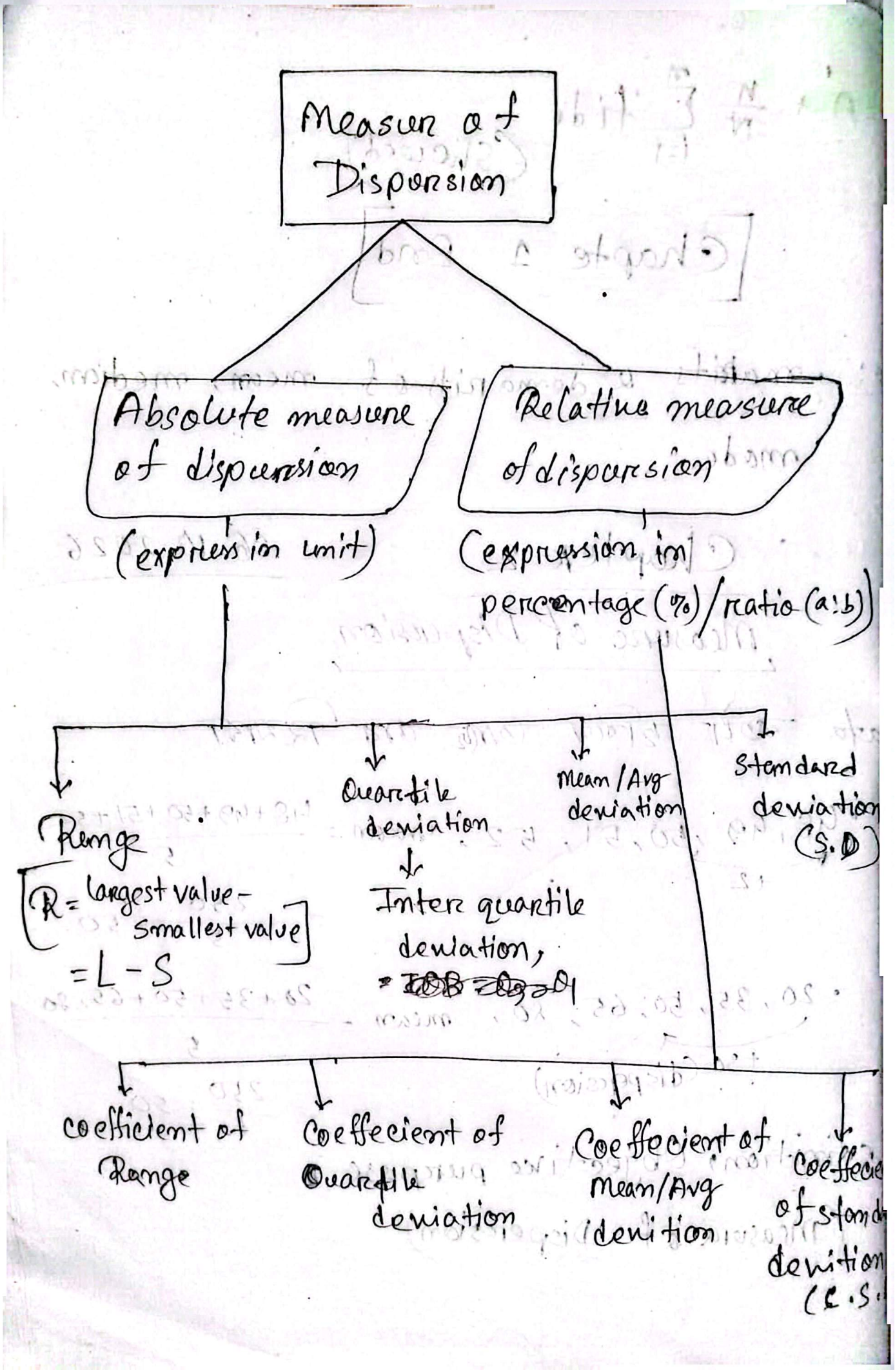
• 20, 35, 50, 65, 80, mean $= \frac{20+35+50+65,80}{5}$

+30 (dispersion)

$= \frac{250}{5} = 50$

⊞ Definition, Objective purpose

(measure of Dispersion)


Measure of Dispersion
Absolute measure of dispersion
(express in unit)
Relative measure of dispersion
(expression in percentage (%)/ratio (a:b))
Range
R = largest value - smallest value
= L - S
Quartile deviation
Inter quartile deviation,
Mean/Avg deviation
Standard deviation (S.D)
coefficient of Range
Coeffecient of Quartile deviation
Coeffecient of mean/Avg deviation
Coeffecie of stand deviation (C.S.

(*) Inter Quartile deviation:

$$IQD = Q_3 - Q_1$$

Quartile deviation:

$$QD = \frac{IQD}{2} = \frac{Q_3 - Q_1}{2}$$

- $Q_1 = \frac{n+1}{2}$ th observation

$$Q_3 = \frac{3(n+1)}{2} \text{ th observation}$$